Local goodness-of-fit measures for neural decoding

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Abstract

State-space models are widely used in neuroscience to infer latent states from observations, but validating them remains difficult. Interpretable models are by nature simplified relative to the data, but it is challenging to determine the level of model complexity needed to accurately describe latent variables. This challenge results from two main factors. First, because latent variables are inherently unobservable, the quality of their decoding cannot be directly validated against observed data. Second, it is often challenging to unravel prediction errors that result from model misspecification. We therefore developed three classes of metrics designed to identify localized periods where model misspecification produces inconsistencies between individual observations and the model's accumulated estimate. Each metric assesses the consistency between the information contained in individual observations and the information extracted by the state-space model from a local window of data. These metrics fall into three categories: 1) a set of divergence measures, specifically f-divergences, which compare observation vs. model distributions of the latent state process directly; 2) measures of overlap between estimated credible regions obtained from the state distributions; and 3) predictive checks for individual observations, based on samples drawn from the assumed state-space model. We applied these metrics to simulated data and to real rat hippocampal neural data, demonstrating their ability to identify periods of model misspecification and to provide interpretable feedback for model improvement. These metrics offer a practical approach for neuroscientists to assess the local goodness-of-fit of state-space models, enabling more informed decisions about model refinement and interpretation of neural decoding results.

Keywords: Neural decoding, State-space models, Goodness-of-fit

Introduction

Internal brain states such as those related to memory, emotion, and attention play a central role in all aspects of behavior, and are thus critical to understand. They arise from an interplay between the external world and internal processes that changes over time, and are not directly experimentally observable. This makes these latent states challenging to understand.

Recordings of neural activity, including spiking, calcium, and voltage traces, can provide experimental measurements that arise as a result of these latent states, and state space models make it possible to relate these measurements to estimates of the latent states themselves. Specifically, these models provide a method for relating complex, high-dimensional neural activity to low-dimensional latent states, and this approach has been deployed to provide insights into neural representations and dynamics related to multiple latent variables, including taste [1], spatial location [2], and aggression [3].

State space models aggregate information over time about the latent states, combining the information from the data at the current time (the normalized likelihood distribution) with the accumulated information used to predict the latent state (the prediction distribution). This powerful combination enables robust estimation of the latent state, effectively balancing current information with the accumulated history of the system and prior knowledge.

Despite the wide use of state space models in neuroscience, methods to validate these models remain underdeveloped and underutilized. One reason is that latent states are inherently difficult to evaluate [4]. Given the absence of ground truth, model validity can only be inferred indirectly, such as from comparing relative likelihoods under multiple models, or from comparing the quality of reconstructed signals, or from comparing performance on downstream tasks. Another challenge is that neural data are dynamic, so model fit may vary substantially across time. Compounding this problem is that common methods of validating models, such as cross-validation and "shuffle" permutation tests, are holistic evaluations of the model — they tell neuroscientists whether or not the model generalizes well or is different from a simplified model with less utility — but they fail to indicate when the model fits poorly or which components are responsible. Consequently, these methods can reject models that are intentionally simplified yet scientifically meaningful and offer little guidance for targeted improvement of the model. For example, analyses to decode rat movements from hippocampal place cells typically ignore known features of the rat's movement (such as momentum) and known features of neurons (such as refractoriness and bursting), which leads to model mismatch according to many goodness-of-fit metrics despite providing reliable and meaningful decoding trajectories [5–7].

State space models already contain rich diagnostic information in the form of distributions used to estimate the state process at each time, including the the normalized likelihood, which reflects information from only the current neural observations, and prediction distribution, which aggregates information from all past observations based on the state space model structure. One potential solution for more targeted local diagnostic information would be directly compare these distriutions. When these two distributions agree, the model's prior expectations and data-driven evidence are consistent. When they diverge, the mismatch can reveal where, when, and potentially how the model fails to capture the structure of the data. Examining these discrepancies could therefore provide a direct, interpretable measure of model–data agreement—analogous to a local, time-resolved goodness-of-fit test.

To address these gaps, we develop and validate a suite of three complementary goodness-of-fit methods – KL divergence, highest posterior density (HPD) region overlap, and predictive checks – designed to provide actionable feedback to neuroscientists. Specifically, we use these methods, each built on the mismatch between prediction and likelihood, to identify time periods of poor model–data agreement and reveal how the three metrics respond selectively to different types of misspecification. We demonstrate the utility of these techniques on simulated and real hippocampal data, offering a path for neuroscientists to move from binary model rejection (is this model good or bad?) toward an iterative, diagnostic approach to building interpretable theories of neural dynamics.

Methods

State-space models

The state-space paradigm is particularly well suited for neural data analysis since many neuroscience experiments focus on internal brain processes that are not directly observable. For example, hippocampal replay of spatial trajectories has been hypothesized to be related to mental exploration of space, but the specific trajectories being replayed are not directly observable by experimentalists [6]. In such cases, state-space models are used to estimate, simultaneously, the unobserved internal representations and their relation to neural activity patterns.

For a discrete set of time points, $t_0, \dots, t_K$, we define an unobserved state process x_k, which influences a neural observation process y_k at each time t_k. We define a state space model by combining a state transition distribution $p(x_k \mid x_{k-1})$, which defines how the state evolves in time, with an observation likelihood $p(y_k \mid x_k)$, which defines the relationship between the neural activity and the state. To simplify the problem, we assume that the state transition distribution depends only on the most recent past value of the state and that the observation likelihood depends only on the current state value. Fig 1(a) shows a schematic of this state-space model structure.

This state-space model structure allows for iterative estimation of the state process at each

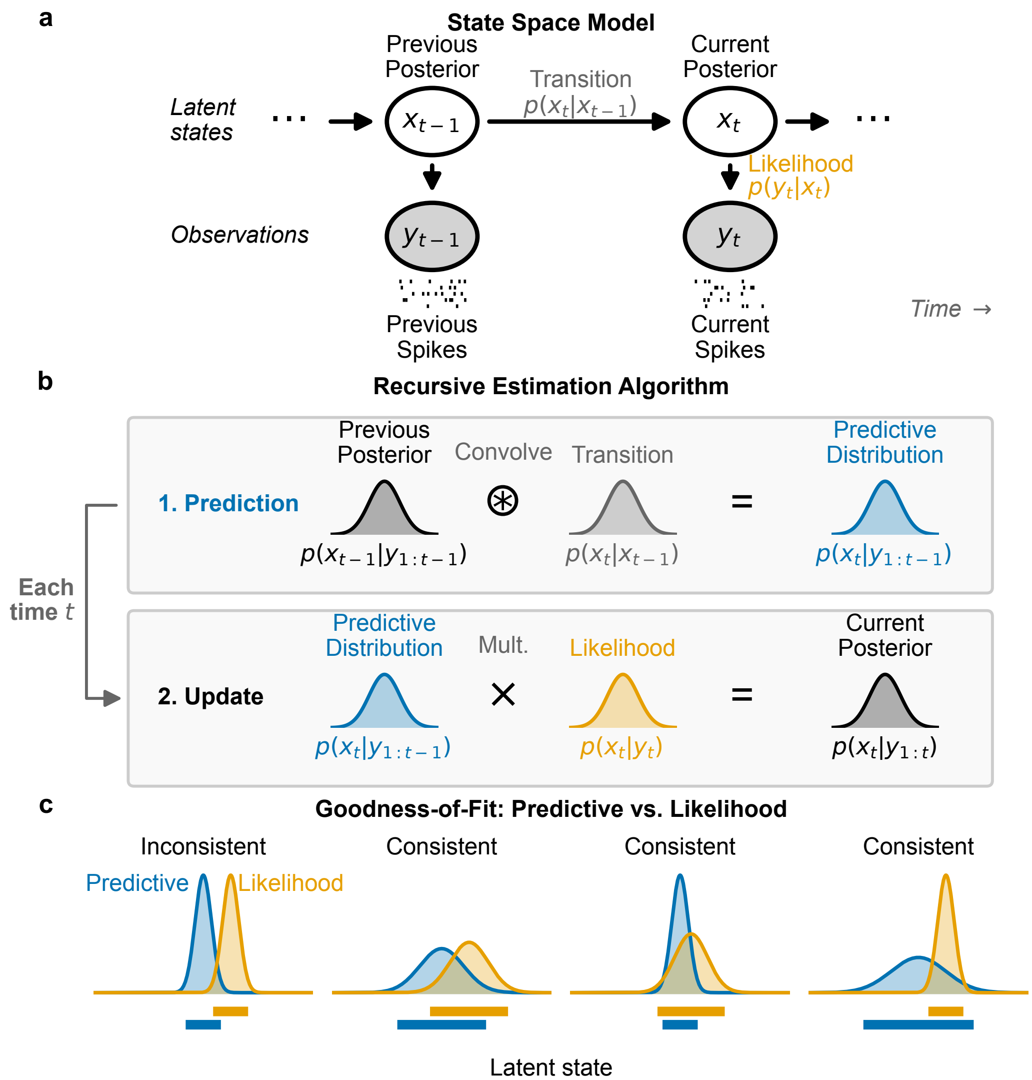

Fig 1. Schematics for state-space model and goodness-of-fit measures. (a) Graphical model depiction of state-space model includes state transition model, $p(x_k \mid x_{k-1})$, and observation likelihood $p(y_k \mid x_k)$. (b) These model components are sufficient to compute an iterative expression for the posterior filter distribution at time t_k, $p(x_k \mid y_{1:k})$ from the distribution at the previous time step. This involves first convolving the previous posterior with the state transition distribution to compute the one-step predictive distribution, $p(x_k \mid y_{1:k-1})$, and then updating the predictive distribution with information from the most recent spiking by multiplying by the observation likelihood. (c) For each new spike observation, we compute the local goodness-of-fit by assessing the consistency of information between the predictive distribution and likelihood. These are considered inconsistent when they assign high probability to mostly disjoint subsets of the state-space. When the regions with high probability overlap, the distributions are considered consistent, even if the distributions differ in their central tendency or certainty.

time step based on all the neural data up to that time (Fig 1(b)). If the distribution of the state at time t_{k-1} has been computed, we can convolve it with the state transition distribution to obtain a one-step prediction distribution, $p(x_k \mid y_{1:k-1}) = \int_{x_{k-1}} p(x_k \mid x_{k-1}) \cdot p(x_{k-1} \mid y_{1:k-1})\, dx_{k-1}$. This predictive distribution combines the information from all previous observations to compute the distribution of the state at the next time step. We update this predictive distribution with the information from the most recent observation by multiplying by the observation likelihood to compute the posterior distribution of the state given all of the data up to the current time step: $p(x_k \mid y_{1:k}) \propto p(y_k \mid x_k) \cdot p(x_k \mid y_{1:k-1})$.

Common approaches for assessing the goodness-of-fit of state-space models include examining the cross-validated error measures in the state process when the state is observable [8], cross-validated error measures in the observation process such as the mean-squared prediction error [9], and the distribution of prediction residuals [10]. While these approaches can be useful, in practice they do not yield easily interpretable assessments for neural data because our state-space models are intentionally simplified and these standard goodness-of-fit measures trivially show lack of fit.

Here we take a different approach that relies on a set of local goodness-of-fit measures, each of which compares the prediction distribution to the normalized likelihood of the most recent observation as a function of the state or to that observation itself. Thus, these measures consider the consistency between information accumulated through the state-space model over past spiking and the information from the most recent spiking activity. Critically, we do not expect these distributions to align perfectly since they likely contain different amounts of information about the state process. Instead, we assess whether the information these distributions provide about the state and observation processes is consistent in the sense illustrated in Fig 1(c). Specifically, we say that the two distributions are inconsistent when their probability mass is concentrated in mostly non-overlapping regions of the state space. This suggests disagreement between the information from past spiking and the information from the most recent spiking. Distributions that largely align, or that are shifted from each other but are supported over largely overlapping regions of state space are considered consistent. Additionally, if one distribution covers a broad region of the state-space and the other covers a much narrower subset of that region, the distributions are considered to be consistent. We expect this situation to arise often since the prediction distribution contains information accumulated from many past spikes.

Local goodness-of-fit measures

In this section, we introduce three classes of metrics designed to characterize the local goodness-of-fit of the state-space model.

Overlap between HPD regions

One way to assess the consistency between the predictive distribution and the observation likelihood is to examine the extent to which their highest posterior density (HPD) regions overlap. A substantial intersection suggests that the two distributions convey consistent information, reinforcing the reliability of the inferred latent state. In contrast, minimal or no overlap signals a significant discrepancy, hinting at potential inconsistencies in the local fit of the model. This approach provides an interpretable and intuitive metric of the degree of consistency between these distributions.

We define the overlap between the HPD for the predictive distribution and the likelihood as

$$\mathrm{HPDOverlap}(P, Q) = \frac{\left|\mathrm{HPD}_P \cap \mathrm{HPD}_Q\right|}{\min\left\{\left|\mathrm{HPD}_P\right|, \left|\mathrm{HPD}_Q\right|\right\}}, \tag{1}$$

where HPD_P and HPD_Q are the highest posterior density regions of the predictive density and the normalized observation likelihood respectively, and $|\cdot|$ denotes the volume of the region. The numerator is the size of the overlap between these regions and the denominator is the size of the smaller of the two HPD regions. This normalization ensures that the resulting value is in the range $[0, 1]$, with 0 indicating no overlap, and 1 indicating that one of the HPDs is entirely contained within the other. This means that if, for example, the likelihood produces an HPD that is wider and therefore less informative than the predictive density but that completely contains the HPD of the predictive density then the HPD ratio will be 1, suggesting that the two distributions are statistically consistent. Conversely, if the ratio is close to 0, it indicates little or no overlap, suggesting significant differences between the range of values for the state that would be inferred from these two distributions. Throughout, we compute 95% HPD regions.

Fig 2(a) illustrates the computation of the HPD overlap. The HPD for the predictive distribution (top) and the normalized likelihood (middle) are computed, and the size of their intersection is normalized by the size of the smaller of the two HPDs (bottom).

Predictive checks

The HPD overlap compared two distributions of the state: one based on the accumulated past information and the other based on the isolated information from the likelihood of the most recent spiking observation. Another approach is predictive checking, which compares the observation itself to a predictive distribution of that observation estimated from the model. This approach has been used before in other contexts ([14–16]) and has the advantage that it allow for the computation of p-values for individual observations, which are easy to interpret [17, 18].

Predictive checks for spiking data can be challenging to implement, since in sufficiently small intervals, any spikes are unlikely compared to no spiking at all. One approach to deal with this might be to consider patterns of spikes over longer fixed intervals over which multiple spikes would be expected. Here, we take a simpler approach of focusing only on spike times and measuring the likelihood that an observed spike comes from a particular neuron (for spike-sorted models) or has a particular set of waveform features (for clusterless models).

In particular, let $y_{k,j}$ be the identity of the neuron (spike-sorted model) or a spike waveform feature (clusterless model) for the jth spike occurring in the kth time bin. We define the predictive distribution of the observation based on the state model as

$$f_{\text{pred}}(y_{k,j}) = \int p(y_{k,j} \mid x_k)\, p(x_k \mid y_{1:k-1})\, dx_k, \tag{2}$$

where $p(y_{k,j} \mid x_k)$ is the observation process and $p(x_k \mid y_{1:k-1})$ is the one-step prediction distribution of the current state given all previous spikes. We can sample new observations $\widetilde{y}_{k,j}$ from this predictive distribution, and compute the probability that the observed spike is at least as likely as the sampled spike:

$$p_{k,j} = \Pr_{\widetilde{y}_{k,j} \sim f_{\mathrm{pred}}} \left(\log f_{\mathrm{pred}}(y_{k,j}) \geq \log f_{\mathrm{pred}}(\widetilde{y}_{k,j}) \right) . \tag{3}$$

The log terms in Eq (4) are unnecessary, but suggest the method to estimate $p_{k,j}$. We sample x_k multiple times from the one-step prediction distribution, $p(x_k \mid y_{1:k-1})$, and use these to generate samples of the spike observations, $\widetilde{y}_{k,j}$, from the predictive distribution. We then compute the fraction of these samples whose log likelihood is smaller than or equal to that of the observed spike. For the spike-sorted models considered here, $\widetilde{y}_{k,j}$ takes one of a finite set of values—the identities of the recorded neurons—so $p_{k,j}$ can be computed exactly by summing f_{pred} over all cell outcomes at least as unlikely as the observed spike, rather than by sampling. Because $p_{k,j}$ ranks the observed spike against this predictive distribution of possible spikes, we refer to it as the *rank-based predictive p-value*.

Fig 2(b) illustrates the computation of a predictive p-value for spike-sorted data. Many values of the state are sampled from the predictive distribution (top), each of which is used to sample a neuron that could have generated the observed spike (middle). The log likelihood of the observed spike is then compared to an empirical distribution of the log likelihoods of the sampled spikes (bottom) leading to a goodness-of-fit metric that can be interpreted as a p-value for that individual spike. High p-values suggest that the observed spike is consistent with the predictive distribution of spikes and a p-value near zero suggests that the observed spike is unexpected based on the predictive distribution computed from the past spiking and the model.

KL divergence

Finally, one of the most well-known measures of discrepancy between probability distributions is the Kullback–Leibler (KL) divergence [11], defined in this case as the expected excess self-information (log of the density function) incurred when the likelihood is used to approximate the predictive distribution, with the expectation taken with respect to the predictive distribution,

$$D_{\mathrm{KL}}(P \,\|\, Q) = \int P(x_k) \log\left(\frac{P(x_k)}{Q(x_k)}\right) dx_k, \tag{4}$$

where $P(x_k) = p(x_k \mid y_{1:k-1})$ is the predictive distribution and $Q(x_k) = p(y_k \mid x_k)/\int p(y_k \mid x_k)\, dx_k$ is the normalized likelihood in x_k given the observed spiking y_k.

Fig 2(c) illustrates the computation of the KL divergence. The top panel shows the predictive distribution and normalized observation likelihood as functions of x_k. The middle panel shows the difference in self-information between these distributions, $\log(P(x_k)/Q(x_k))$. The bottom panel shows this difference in self-information weighted by $P(x_k)$, which when integrated gives the value of the KL divergence.

The KL divergence is an asymmetric measure, meaning that $D_{\mathrm{KL}}(P \,\|\, Q) \neq D_{\mathrm{KL}}(Q \,\|\, P)$. We could also compute the KL divergence from the likelihood to the predictive distribution $D_{\mathrm{KL}}(Q \,\|\, P)$ or the average of these two divergences, $\frac{1}{2}(D_{\mathrm{KL}}(P \,\|\, Q) + D_{\mathrm{KL}}(Q \,\|\, P))$, but neither of these is recommended as the KL divergence can become arbitrarily large when the first distribution has a broader support than the second, which would occur whenever an uninformative spike produces a broad likelihood.

Simulation study

Validating local goodness-of-fit metrics on real neural data is fundamentally limited: when the truth is unobserved, a metric that flags a moment of model failure cannot be distinguished from one that flags a moment of unusual neural activity. We therefore first evaluate the three diagnostics—KL divergence, HPD overlap, and the rank-based predictive p-value—on a controlled benchmark in which the ground-truth fault at each moment is known by construction. The benchmark simulates a hippocampal place-cell population on a one-dimensional track and runs a Bayesian state-space decoder over the full session, deliberately perturbing either the generative process or the decoder's assumptions during four distinct windows. These four perturbations were chosen to highlight the

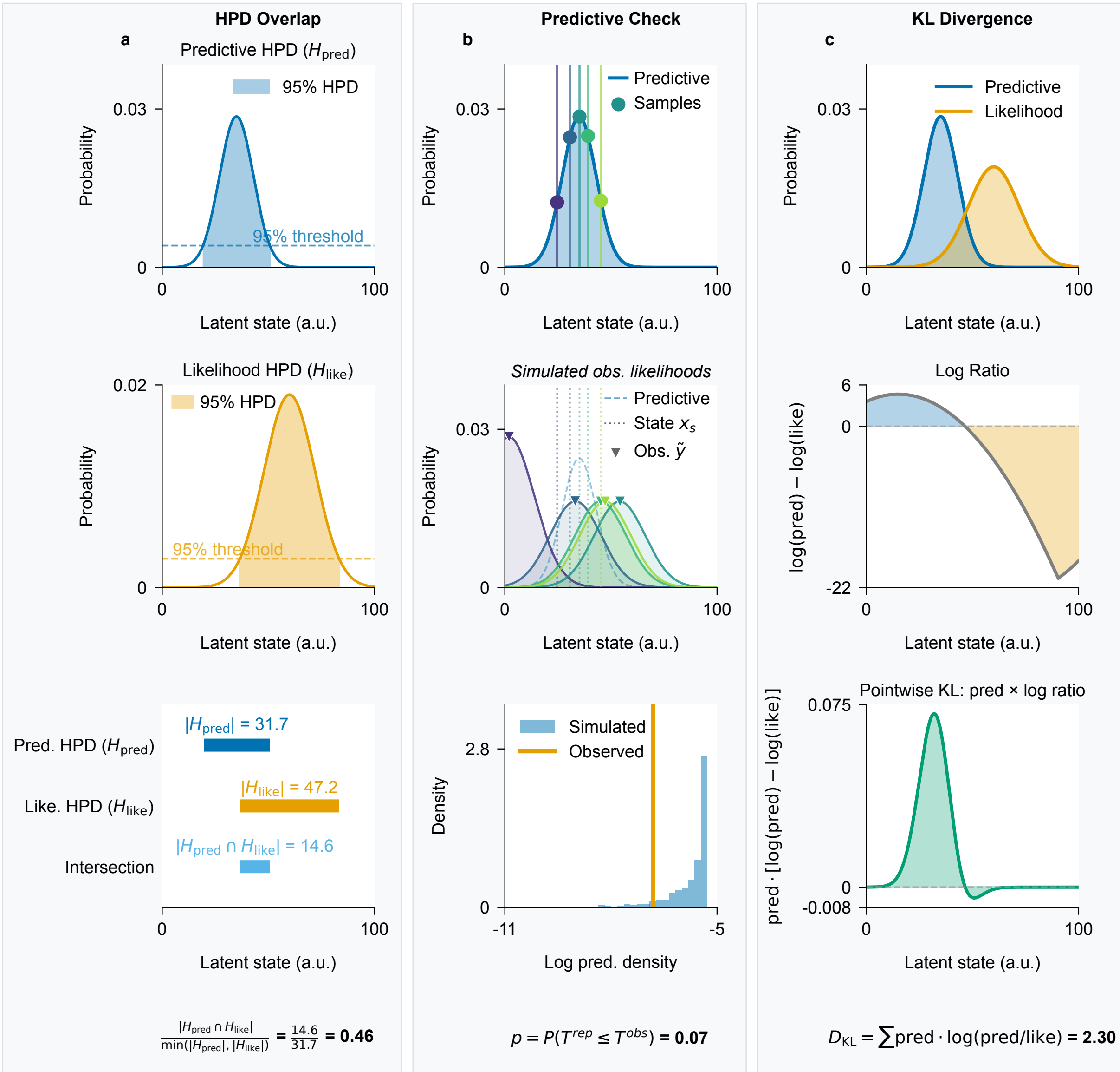

Fig 2. Visualizations of the computation of each local goodness-of-fit metric. (a) The KL divergence compares the predictive distribution (blue) and the observation likelihood (orange) (top panel) by computing the difference in their log density (middle panel) and computing its expectation with respect to the predictive distribution (bottom panel). (b) The HPD overlap is defined by computing the highest posterior density (HPD) for the predictive distribution (top) and for the observation likelihood (middle) and then computing the size of their intersection relative to the size of the smaller of these two HPD regions (bottom). (c) Predictive checks are computed by sampling values of the state from the predictive distribution multiple times (top), then sampling from the conditional distribution of neurons that could have generated the observed spike given each sampled state value (middle), and then computing the probability that the log of the predictive distribution of the observed spike is greater than the log of empirical predictive distribution of the sampled spikes (bottom).

distinction between model misfit that leads to decoding errors versus model misfit to which the decoding estimates remain robust. This relates to our original goal of identifying local periods where the information from the current spiking is inconsistent with the aggregated information from our model based on past spiking.

Generative model

The simulation runs a Bayesian filter over a 32-second session whose generative process and decoder coincide everywhere except inside four misfit windows. Each window perturbs one specific component of the matched model.

Track and observation model. The state space is a one-dimensional linear track of length $L = 100$ arbitrary units (a.u.) discretized at unit spacing, $x \in \{0, 1, \ldots, 100\}$. The simulated population consists of eleven place cells with Gaussian place fields centered at $\mu_c \in \{0, 10, \ldots, 100\}$ and shared standard deviation $\sigma_{\text{pf}} = 10$ a.u. The expected spike rate of cell c at position x is

$$\lambda_c(x) = \alpha\, \phi\Big(x \mid \mu_c, \sigma_{\text{pf}}^2\Big), \tag{5}$$

where $\phi(\cdot \mid \mu, \sigma^2)$ is the Gaussian density and $\alpha = 5$ is a peak-rate scale factor. With a 1 ms simulation step, the peak rate is $\alpha/(\sigma_{\text{pf}}\sqrt{2\pi}) \approx 200\,\text{Hz}$. Spike counts are drawn independently across cells at each step, $y_{c,t} \sim \text{Poisson}(\lambda_c(x_t))$.

Position dynamics. In baseline windows the animal's position evolves as a Gaussian random walk with step standard deviation $\sigma_{\text{pred}} = 0.5$ a.u. The trajectory is reflected at 0 and 100 a.u. throughout the session, starts at $x_0 = 0$, and is continuous across phases: the final position of each phase becomes the initial position of the next.

Phase schedule. The 32-second simulation ($T = 32{,}000$ steps) is partitioned into eight contiguous phases: an opening baseline, four misfit windows, and three clean-recovery interludes that restore the matched-model configuration. Each misfit window perturbs one component of the model—the observation model $p(y_k \mid x_k)$ or the state-transition model $p(x_k \mid x_{k-1})$—by altering either the generative process or the decoder's assumptions:

Remap: During steps 6,000–10,000, the decoder-side observation model is perturbed. The decoder evaluates per-spike likelihoods against a remapped set of place-fields for which three cell pairs ($0 \leftrightarrow 9$, $1 \leftrightarrow 8$, $2 \leftrightarrow 7$) have swapped centers, so the decoder's intended likelihood is at the wrong location for six of the eleven cells. Spike generation is unchanged.

History-dependent firing: During steps 14,000–18,000, the generative-side observation model is perturbed. Spiking is generated with a 1 ms hard refractory period followed by a 2–10 ms burst window with a threefold rate increase, matching the rough phenomenology of CA1 pyramidal-cell refractory and burst windows; the decoder still assumes independent Poisson firing. The per-spike spatial likelihood is unchanged, so the misfit lives in the spike-train temporal joint distribution.

Drift: During steps 22,000–26,000, the generative-side state-transition model is perturbed. The trajectory is generated with persistent velocity, $v_t = 0.88\, v_{t-1} + \eta_t$ and $x_t = x_{t-1} + v_t$ with $\eta_t \sim \mathcal{N}(0, \sigma_{\text{pred}}^2)$; the decoder continues to assume a memoryless random walk with step standard deviation σ_{pred}.

Wide-dynamics noise: During steps 30,000–32,000, the decoder-side state-transition model is perturbed. The decoder uses an inflated transition matrix with step standard deviation $\sigma_{\text{inflated}} = 20$ a.u. (forty times baseline), so its one-step predictive distribution spreads broadly across the track; spike generation is unchanged. This phase is engineered to inflate KL divergence while leaving HPD overlap and the rank-based predictive p-value near baseline.

The three intermediate clean-recovery windows (steps 10,000–14,000; 18,000–22,000; 26,000–30,000) restore both the generative process and the decoder to the baseline configuration, providing within-session reference periods against which each metric's behavior during a misfit can be calibrated. With this schedule fixed, each diagnostic can be evaluated directly against a known ground-truth fault, and the three responses can be compared across faults that differ in which component of the model is perturbed.

Simulation results

Fig 3a shows a realization of the simulation, the decoding results, and each of the three local goodness-of-fit metrics. The top panel shows the true position of the simulated rat in magenta and the predictive decoding distribution in black. The regions with white backgrounds are the periods when the model is correctly specified and each colored background indicates a period of model misspecification. The second panel shows the normalized likelihood of the data as a function of position at the spike times. The third panel shows the spiking from each of the eleven simulated neurons. Since these are sorted by their place field centers, the pattern of spiking aligns with the likelihood as a function of position. Note that the remapped period is the one in which the predictive distribution and the spike likelihoods most clearly disagree about the animal's location. In the other misspecified periods this disagreement is weaker and, as detailed below, surfaces unevenly across the three metrics.

The lower three panels of Fig 3a show, from top to bottom, the values of the HPD overlap, the rank-based predictive p-value (as $-\log(p)$) and KL divergence for each spike. In each panel a horizontal line marks the threshold used to flag poor fit. For the HPD overlap, where low values indicate misfit, a spike is flagged when its value falls below the threshold, taken as the 1st percentile of the HPD overlap over the baseline periods pooled across 100 independent realizations of the simulation. For the KL divergence, where high values indicate misfit, a spike is flagged when its value exceeds the threshold, taken as the 99th percentile over the same pooled baseline. Pooling many realizations stabilizes these baseline-derived thresholds, which are otherwise noisy when estimated from a single run. For the predictive p-value, a spike is flagged when $p \leq 0.05$ (equivalently $-\log(p) \gtrsim 3$), a fixed cutoff rather than a baseline-derived bound. We see that all three metrics agree that there is lack of consistency during the remapping period, where the

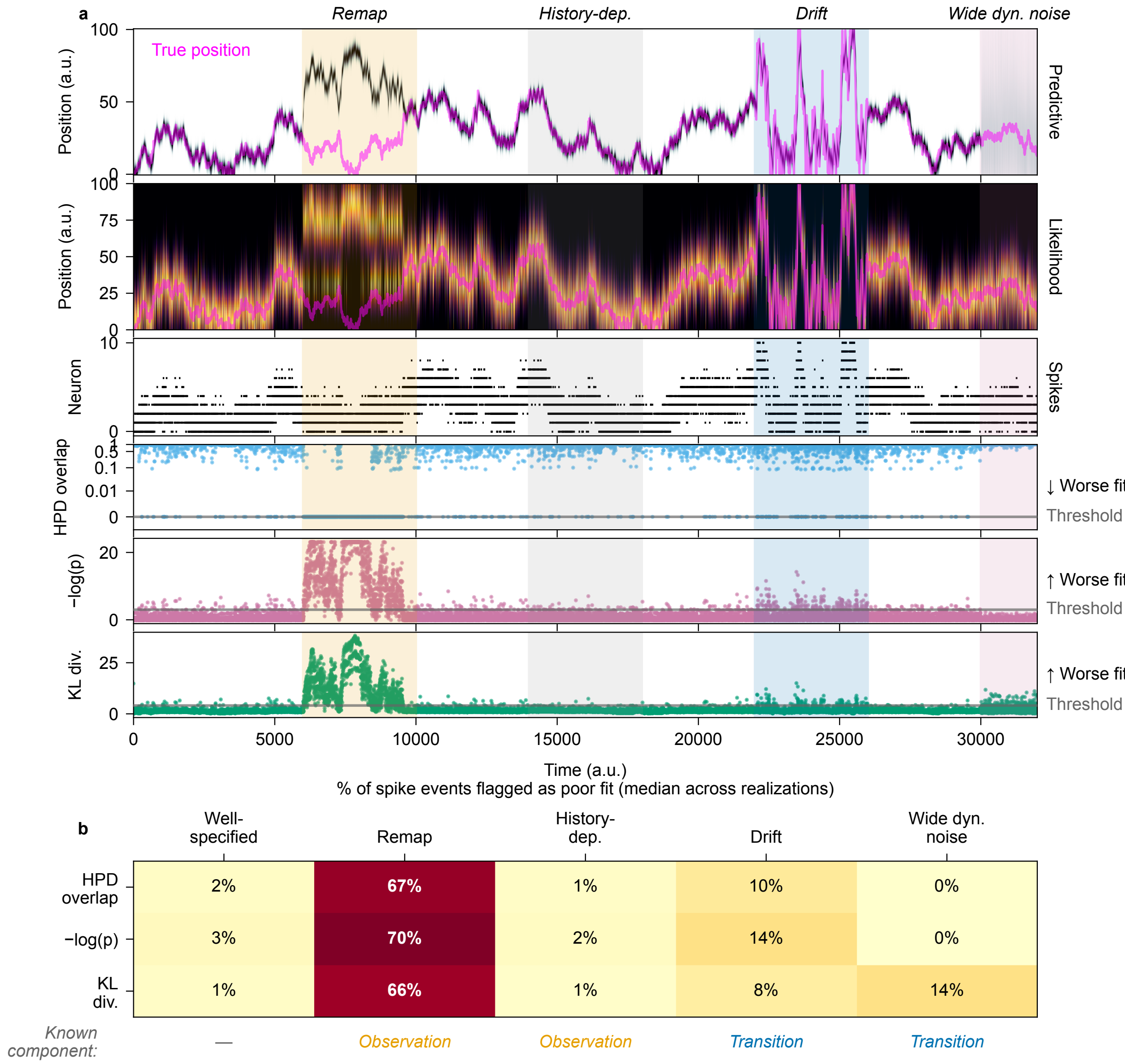

Fig 3. Per-spike goodness-of-fit diagnostics across a four-misfit simulation. The simulation walks through four model misspecifications separated by clean-recovery windows: place-field remapping, unmodeled history-dependent firing, persistent-velocity drift, and wide-dynamics noise. (a) The top block shows the predictive distribution as a heatmap of time and position. The position is overlaid as a magenta line. The misfit time periods are shaded in color. The second from the top block shows the per-spike likelihood as a heatmap, restricted to time bins containing at least one spike. The position is also overlaid as a magenta line. The third block shows the spike raster with neurons sorted by place-field peak position. The bottom three blocks show the three goodness-of-fit diagnostics (HPD overlap, KL divergence, and the rank-based predictive p-value) for each spike event, with a horizontal line marking the threshold used to flag poor model fit; the HPD overlap is plotted on a symmetric-log y-axis to resolve values near zero. (b) The percent of spike events flagged as poor fit by each diagnostic in each phase, reported as the median across 100 independent realizations of the simulation, with the well-specified "clean" recovery windows pooled into a single column for comparison. Each realization is scored against shared flag thresholds derived by pooling the baseline periods of all 100 realizations; pooling stabilizes both the thresholds and the per-phase fractions, which are noisy when estimated from any single run. A row beneath the heatmap labels each phase by the model component its misfit perturbs—the observation model (*Observation*) or the state-transition model (*Transition*)—with no induced fault (—) in the well-specified phases.

likelihood from the remapped cells is in a different part of the environment than the model estimate based on the incorrect place field estimates. For the period where history dependence of spiking is misspecified, the estimated decoding distribution becomes narrower, but remains consistent with the spike likelihoods, and none of the three metrics flag this period as leading to incorrect estimates. For the period where there is a drift term that is missed by the decoder model, the actual movement trajectory has notably increased velocity. The decoder only partially keeps up, so its predictive estimates lag behind the true location; this lag biases the predictive spiking distribution toward neurons consistent with the lagged position. As a result, the diagnostics flag a modest fraction of spikes during the drift period—most clearly the rank-based predictive p-value—though far less than during the remapping period. For the period where the decoder uses an inflated transition matrix, the predictive density becomes nearly uniform, losing all information from past spiking. Both the HPD overlap and rank-based predictive p-value measures indicate that this broad predictive distribution is consistent with the likelihood of each spike. Here the model misspecification is identified by the KL divergence identifies the mismatch because the distributions are now inconsistent. This highlights the complementary strengths of the KL divergence and the other two metrics.

Fig 3b shows a heatmap of the fraction of spikes flagged by each metric in each period, summarized as the median across 100 independent realizations of the simulation. A spike is flagged whenever its diagnostic crosses the corresponding threshold in the direction of worse fit: below the 1st-percentile baseline value for the HPD overlap, above the 99th-percentile baseline value for the KL divergence, or at or below the fixed 0.05 cutoff for the predictive p-value. The median fraction flagged during the remapping period is large for all three metrics (76–81%), demonstrating that all three methods are able to identify inconsistent spiking information leading to incorrect state estimates; this fraction nonetheless varies widely from realization to realization because it depends on how often the simulated trajectory enters the remapped fields. All three metrics flag only a few spikes during the history-dependence period; the drift period is flagged a bit more—most notably by the rank-based predictive p-value—but still far less than the remapping period. For the period where the decoding model incorrectly specifies the state transition, and therefore overly discounts previous spike information, the HPD overlap and rank-based predictive p-value metrics do not identify inconsistency between the spike likelihoods and the flat predictive distribution, while the KL divergence flags a median of 14% of the spikes. On one hand, this suggests that the KL divergence is sensitive to this true model misspecification; on the other hand, this suggests that the KL metric will flag spikes whenever the predictive distribution is uninformative, even when that distribution is consistent with the information from the current spike.

Real hippocampal data analysis

Data collection and behavioral task

We examined the ability of these local goodness-of-fit metrics to identify periods of inaccurate decoding from cells in the rat hippocampus as the animal performed a spatial bandit task on a maze with three radially arranged Y-shaped foraging patches emanating from a center point. A schematic of the maze and the rat's movement in this environment are shown in the track inset in Fig 4. The details of this task and the data collection methods are described by Comrie et al. [19]. Briefly, the rat was trained to run between six reward ports located in three Y-shaped "patches", each with a different probability of reward. The probability of reward periodically switched after a time or trial limit. Twenty-three tetrodes were implanted in the dorsal hippocampus CA1 region. Spikes were detected by a 100 µV threshold on the recorded neural signal filtered 600 Hz to 6,000 Hz. Spikes were sorted into putative single units with MountainSort4 [20]. The 30 kHz signal was whitened, and negative-going spikes were detected on each tetrode using a threshold of 3 standard deviations of the whitened signal, and a minimum inter-event interval of 10 samples (≈0.33 ms). Units were retained from automated curation only, without manual cluster curation. Because curation was automated, some units may contain multiunit activity or represent incompletely merged clusters; this does not affect our purpose here, which is to demonstrate the diagnostics rather than to characterize individual neurons. For decoding, we linearized the animal's position by projecting it to the nearest location of a skeletonized track and discretized it into spatial bins of approximately 2 cm, and binned spikes at 2 ms resolution. Because the real recording has no designated clean-baseline period, we flag spikes using fixed cutoffs (HPD overlap and predictive p-value below 0.05) rather than the within-session baseline percentiles used in the simulation.

Data analysis results

Fig 4 shows a comparison of decoding from this hippocampal spiking data under two state models. The first assumes that the state process moves continuously through space at all times, as would be true if the state only represented the rat's actual location. However, it is known that during periods where the rat is immobile, hippocampal place fields can both jump to distant locations and reactivate in sequences that replay past movement trajectories [21, 22]. We expect the continuous state model to fail precisely at those times when the neural representation jumps. We compare this with an alternative state model called a continuous-fragmented model that allows the hippocampal representation to jump at discrete times. We have previously used a version of this model structure to capture replay and other non-local hippocampal representations [2].

Fig 4a shows a brief segment of the decode using the continuous state model during a period when the rat is immobile. The panels from top to bottom are the same as in Fig 3a. While the rat rests (magenta line), we find a classic replay event with rapid transitions between areas of the maze. During this replay event, the continuous state model is unable to keep up and the predictive distribution consistently lags behind the information coming from the likelihood of the observed spikes. In the lower three panels, we see many of the spikes flagged as inconsistent with the predictive model estimates by all three goodness-of-fit metrics.

The contrast between decoding with the continuous and continuous-fragmented state models is shown in Fig 4a and b. We see that the likelihood for each spike is identical for these models (second panel from top) but the continuous-fragmented model decodes a rapidly moving state trajectory that the continuous model is unable to track. This happens because the predictive distribution becomes uniform or nearly uniform at time steps where the representation is estimated to be fragmented, and the likelihood of subsequent spikes can then cause a jump in the state estimate. For the HPD overlap

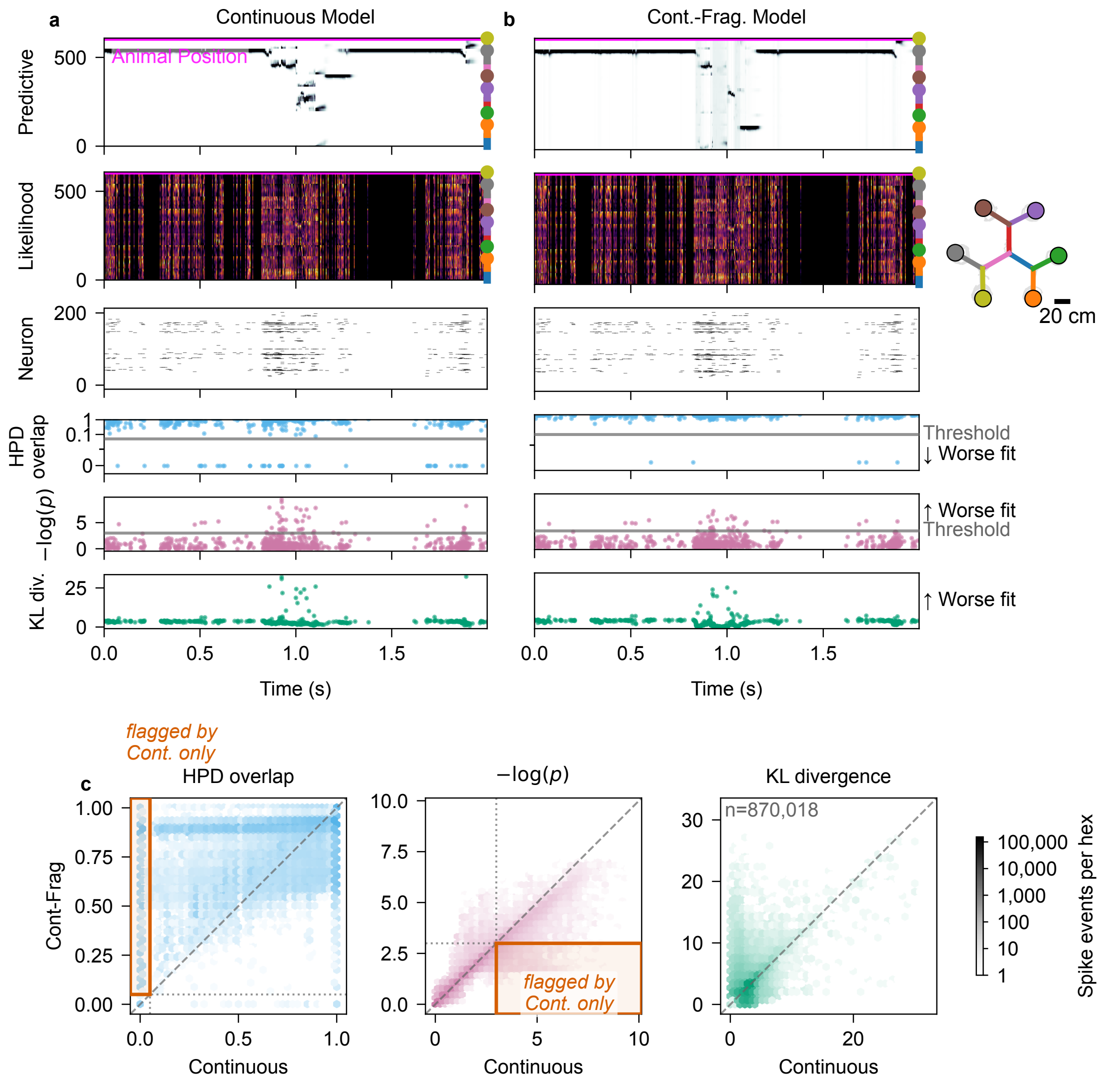

Fig 4. Per-spike goodness-of-fit diagnostics applied to a real hippocampal recording (203 simultaneously recorded cells) during navigation on a multi-arm maze; data from Comrie et al. [19]. Two state-space decoders—a purely Continuous decoder and a Continuous–Fragmented decoder—are evaluated on the same session. The unlettered track inset shows the 2D layout of the track, with the maze arms drawn as colored segments, colored dots marking the six reward wells at the arm ends, the animal's trajectory overlaid in grey, and a 20 cm scale bar. **(a), (b)** A detail window containing a replay event, decoded with the Continuous and Continuous–Fragmented decoders, respectively. Within each of (a) and (b), all rows share a common time axis (x, seconds, relative to the start of the window). **Row 1, Predictive**: heatmap of the predictive posterior with linearized position on y and the animal's true linearized position overlaid. **Row 2, Likelihood**: same axes, restricted to time bins containing at least one spike, showing the combined per-spike likelihood. **Row 3, Neuron**: spike raster with neurons on y, sorted by place-field peak position. **Rows 4–6 (HPD overlap, KL divergence, $-\log(p)$)**: one marker per spike event placed at its spike time, with the per-spike diagnostic value on y; for the HPD overlap and the predictive p-value a horizontal line marks the threshold used to flag poor model fit (a fixed 0.05 cutoff), the KL divergence has no natural fixed cutoff and is shown without a threshold, and the direction of worse fit is labeled at the right. **(c)** Pairwise comparison of per-spike diagnostic values between the two decoders across the whole session, one hexbin panel per metric (HPD overlap, KL divergence, $-\log(p)$); x-axis is the Continuous decoder, y-axis is the Continuous–Fragmented decoder, and the dashed line is identity. For the HPD overlap and $-\log(p)$ panels, dotted lines mark the per-metric flag threshold on each axis (the same 0.05 cutoff used in panels (a)–(b)) and the outlined box marks the quadrant of spikes flagged by the Continuous decoder but not the Continuous–Fragmented decoder; the KL divergence panel has no threshold. Color intensity indicates spike-event count on a shared log scale.

metric (fourth panel from top, blue), we see that nearly all of the spikes flagged in the continuous state model are no longer flagged when using the continuous-fragmented model. For the rank-based predictive p-value (bottom panel, pink), we see that some of the $-\log(p)$ values are reduced for the continuous-fragmented relative to the continuous state model. This is because the continuous state model leads to predictive densities that are concentrated in the wrong location, which makes the observed spiking very unlikely, while the continuous-fragmented state model leads to broad predictive distributions during replay events, which makes the observed spikes still unlikely but less so. For the KL divergence metric (fifth panel from top, green), we find that many spikes retain high values for the continuous-fragmented model. This is consistent with our simulation results, where nearly flat predictive distributions led to increased KL values, even though they are consistent with the spiking likelihood according to our earlier definition.

Fig 4c shows hexbin density plots comparing the values of each of the local goodness-of-fit metrics for the continuous state model versus the continuous-fragmented state model applied to each spike, with each hexagon colored by the number of spike events it contains. The leftmost panel shows the values of the HPD overlap metric in blue. Here, hexagons above the 45° line correspond to spikes for which the continuous-fragmented state model produces predictive distributions more consistent with the spiking data than the continuous state model. Of note here is the dense vertical band at $x = 0$ and $y > 0$ (within the outlined box), corresponding to spikes for which the predictive distribution from the continuous state model has no overlap with the spike likelihood but for which the predictive distribution from the continuous-fragmented state model has nonzero overlap. There is no corresponding density at $y = 0$ and $x > 0$, suggesting that the HPD overlap is successful at identifying times when the continuous state model is inconsistent with the observed spiking and that adding the fragmented state process into the model alleviates this inconsistency. Quantitatively, of the 18,328 spikes that the continuous model flags under the HPD overlap, 17,123 (93%) are no longer flagged by the continuous-fragmented model (which we call *rescued*), while only 139 spikes are flagged by the continuous-fragmented model but not the continuous one (*newly flagged*).

The center panel of Fig 4c shows values of the KL divergence in green. In this case, hexagons below the 45° line correspond to spikes for which the continuous-fragmented state model produces predictive distributions more consistent with the spiking data than the continuous state model. There is indeed a large concentration of density below this line in the region where the value for the continuous model is below 5. More striking though is the region above the 45° line where the value for the continuous model is around 2 and the value for the continuous-fragmented model goes up to 20 or so. As seen in the simulation, this corresponds to situations where the predictive distribution from the continuous-fragmented model is broad, as occurs whenever a fragmented state is estimated. For this reason, the KL divergence is less useful in identifying inconsistency from the continuous-fragmented model.

The right panel of Fig 4c shows the value of the negative log of the rank-based predictive p-value ($-\log(p)$) in pink. Once again, hexagons below the 45° line correspond to spikes for which the continuous-fragmented state model produces predictive distributions more consistent with the spiking data than the continuous state model. Of note here is the band of density below this line that extends beyond a value of 3 on the x-axis (within the outlined box, corresponding to a p-value below 0.05, since $-\log(0.05) \approx 3$). The corresponding region above the line with a value for the continuous-fragmented model above 3 has substantially reduced density. As with the HPD overlap metric, this suggests that the rank-based predictive p-value is also successful at identifying times when the continuous state model is inconsistent with the observed spiking. Quantitatively, the continuous-fragmented model rescues 8,786 of the spikes flagged by the continuous model under the predictive p-value while newly flagging only 1,654.

Discussion

State space modeling has become a powerful tool for describing the latent dynamics of neural systems. However, validating these models is inherently difficult because latent states are unobservable and many models are purposefully simplified. To address this, we developed multiple local goodness-of-fit metrics that compare the instantaneous information received from each spike to the aggregated information computed based on recent spiking activity and the state-space model assumptions. Rather than dismissing a model in its entirety, these metrics are able to assess the consistency between immediate observations and the information accumulated through the state-space structure, providing a more granular assessment of decoding reliability.

We defined three classes of metrics: f-divergence measures (e.g., KL divergence), overlap ratios between credible regions (HPD overlap), and predictive checks. Although KL divergence is computationally simple, it is sensitive to distribution tails and lacks a clear upper bound, making it difficult to assess similarity. In contrast, the HPD overlap offers a more robust geometric measure of similarity, consistently maintaining lower error rates across various conditions. Predictive checks provide a direct statistical assessment through observation-level p-values, but are more computationally intensive. Although each metric has particular advantages, when used in combination, they provide a more complete perspective on decoding reliability.

We illustrated the application of these metrics to simulated and real spike-sorted data from rat hippocampus. These same metrics naturally extend to other neural data types when incorporated into state space models. In particular, recent work has shown that clusterless neural spiking models can substantially reduce bias and variance in neural decoding and estimation problems when compared to spike-sorted models [23]. The metrics, as defined in the Methods section, naturally extend to clusterless models. In that case, a lack of fit for any particular spike suggests that a spike with that waveform was unexpected based on the accumulated information from the state space model up to that point in time.

These metrics make it possible to identify specific periods of time where a state-space model is more or less consistent with the spiking data. For example, we show that a model that includes a fragmented state that allows jumps in hippocampal representations provides a better fit to many spikes than a continuous-only model where these jumps are highly unlikely. More broadly, these approaches enable a feedback loop of model development and model assessment that could greatly improve the quality of our state-space models.

Our approach could also be extended. For simplicity, we focused on comparing instantaneous information in the likelihood of each spike with aggregated information from the past contained in the one-step prediction density. A natural and powerful extension of these methods is obtained by replacing the one-step prediction density with any posterior density derived from the state-space framework. In particular, replacing the one-step prediction density with a smoothing density (e.g., from a fixed-interval smoother over the observation period) would allow us to leverage both past and future data to increase statistical power in identifying model misspecification.

This potential impact of these approaches does come with challenges, however. First, the computational burden of computing these metrics increases as the dimensionality of the state process increases. In all of our examples, we computed the KL divergence measure and the HPD overlap by partitioning the state space into a lattice and computing the likelihoods and prediction densities at each point in that lattice. As the dimension of the state increases, the size of that lattice will grow exponentially. While the predictive checks are the most computationally burdensome in lower state dimensions, the fact that we are sampling from the state-space rather than computing values over a lattice means that in higher dimensions the predictive checks will become more computationally efficient. It also suggests that we may be able to improve the efficiency of the KL and HPD overlap metrics by using sampling-based estimation methods. This will not solve the curse of dimensionality, as the number of samples required for accurate estimates will likely still grow with dimension, but it is likely to allow for efficient estimation in moderate dimensions. Another approach for dealing with this issue may be to compute these consistency metrics in lower-dimensional subspaces of the state space. Identifying subspaces that retain the statistical power of the original space is a topic for future exploration.

Another potential challenge relates to identifying time-scales for which to assess for lack of fit. These methods allow us to focus on time-scales as short as individual spikes, and for our place cell decoding examples individual spikes are sufficiently informative that we are often able to detect local model misfit at this time-scale. For other neural systems, it may be the case that local model errors are more likely to be detected over longer time-scales. In this paper, we noted that these goodness-of-fit measures can be extended to longer time intervals, but determining what time-scales are most appropriate may require relying on prior knowledge about the system, or exploring the values of these metrics across multiple time-scales.

We believe that these local goodness-of-fit tools provide an important step toward making inferences from state-space neural models more reliable and interpretable. By pinpointing specific moments when the model assumptions break down, researchers can be more confident in their inferences. Ultimately, these measures empower us to refine our understanding of neural dynamics and deepen our understanding of how the brain encodes and processes information.

Data availability

[TODO: Add accession numbers, repository names, and persistent URLs/DOIs for any raw or processed data required to reproduce the results.] The hippocampal recording analyzed here was originally reported by Comrie et al. [19].

Code availability

All simulation, analysis, and figure-generation code is publicly available at `https://github.com/edeno/statespacecheck-paper`. [TODO: add the archival DOI (e.g. Zenodo) for the tagged release used in this paper.]

Ethics statement

[TODO: Add animal ethics approval information for the rat hippocampal recording, including the approving institutional committee and protocol or permit number.] All animal procedures were approved by the Institutional Animal Care and Use Committee at University of California, San Francisco.

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[TODO: Add all funding sources and grant numbers. State whether the funders had any role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.]

Competing interests

The authors have declared that no competing interests exist.

Author contributions

Conceptualization: UTE, SZ **Methodology:** UTE, SZ **Software:** SZ, ELD **Formal analysis:** SZ, ELD **Investigation:** SZ, UTE, ELD **Data curation:** AEC **Writing – original draft:** SZ, UT, ELD **Writing – review & editing:** SZ, AEC, LMF, UTE, ELD **Visualization:** SZ, ELD **Supervision:** UTE **Funding acquisition:** UTE, LMF

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